

SPSS Activity: Data Preparation, Visualization, and Analysis

Analysis Scenario: Dog Toy Durability & Preference Study

A pet products company wants to design better toys for dogs. To understand toy preferences and durability, they collected data from dog owners. Each owner reported:

- Their dog's breed and size
- How many toys their dog already owns
- The dog's favorite toy
- A toy the dog chose during a product test
- How long the toy lasted before breaking (DaysToFail)

Research Questions:

1. **Toy Preference**
 - Which toy types (Duck, Hedge Hog, Lamb Chop) are chosen most often?
 - Do toy preferences differ by dog breed or size?
2. **Durability**
 - Which toys last the longest before failing?
 - Does durability differ across dog breeds or sizes?
3. **Dog Behavior Insight**
 - Do dogs with more toys tend to favor more durable toys?
 - Are certain breeds more destructive (shorter DaysToFail)?

Instructions:

- Import and prepare the dataset
- Analyze the data using appropriate methods
- Export all your SPSS outputs (tables, charts, and analyses) in APA format to a Word document
- Interpret any notable pattern and extract **insights** to support the decision-making process

Q 1 – Import and Define Variables

- Import the provided dataset into SPSS.
- Ensure that all variables are correctly defined (e.g., numeric, string, nominal, ordinal).

Q 2 – Data Cleaning and Preparation

- Check the frequency table for each variable
- Inspect your dataset for duplicates, inconsistencies, and missing values.
- Identify and fix any leading or trailing spaces in text fields.

Q 3 – Descriptive Statistics

- Compute the following descriptive statistics for your numeric variables:
Mean, Median, Mode, Standard Deviation, Variance, Standard Error of the Mean (S.E),
Quartiles (Q1, Q2, Q3), Minimum, Maximum, Interquartile Range (IQR), and
Frequencies.
- Present these table results neatly in your report in APA 12 pts times new roman style.

Q 4 – Outlier Identification

- Determine manually in your report the outlier interval using the IQR rule:
Lower bound = $Q1 - 1.5 \times IQR$
Upper bound = $Q3 + 1.5 \times IQR$

Q 5 – Boxplot (DaysToFail Variable)

- Use SPSS to construct a boxplot for the “DaysToFail” variable.
- Identify and inspect if any outliers are present, and state how are you going to deal with it in your report.
- Change the box color to light grey to maintain APA visual style.

Q 6 – Filtered Visualization

- Create another boxplot for the “DaysToFail” variable, but filter out small dog sizes before generating it.
- Comment on whether the results change compared to the full dataset.

Q 7 – Comparative Boxplot

- Create a boxplot of DaysToFail for each ToyChosen type to compare durability. Which toy appears to be the most durable, and which one fails the fastest? and do you see any outliers?
 - What about dog size? Do DaysToFail differ for each dog size? Create a boxplot to compare
 - What about dog breed? Do DaysToFail differ for each dog size? Create a boxplot to compare
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Q 8 – Histograms

- Create histograms for both DaysToFail and ToysOwned variables.
 - Apply the following settings:
 - Change bar color to light grey.
 - Remove gridlines.
 - Set Y-axis minimum value to 0.
 - Add a density curve.
 - Determine whether the distribution appears normal or skewed and describe this in your report.
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Q 09 –Pie Chart & bar chart

- Identify the most chosen toy by creating a pie chart for FavoriteToy, display the Percentage, Count, and the name of the toy on the pie chart slices
 - Identify the most chosen toy by creating a bar chart count for FavoriteToy, display the count and percentage on each bar for each toy
 - Compare the results of bullet 1 and bullet 2, which one is better to use to present our data analysis?
 - Create a Clustered Bar Count of FavoriteToys by DogBreed to see if toy preference varies by breed
 - Create a Stacked Bar Count of FavoriteToys by DogBreed, compare the result with the clustered bar count, which one is better?
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Q 10 – Scatterplot with Regression Line

- Create a scatterplot of DaysToFail vs. ToysOwned.
 - Add a fit line (linear trend).
 - Customize colors to light grey and remove gridlines for APA compliance.
 - Interpret the pattern: Is the relationship positive or negative?
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Q 11 – Correlation Analysis (Do dogs with many toys destroy them faster?)

- Calculate the correlation between DaysToFail and ToysOwned
- Discuss:
 - Is there a correlation?
 - Is it statistically significant? What does statistically significant mean?
 - Compare your correlation calculation results with the scatter plot graph for the same variables? What did you notice?